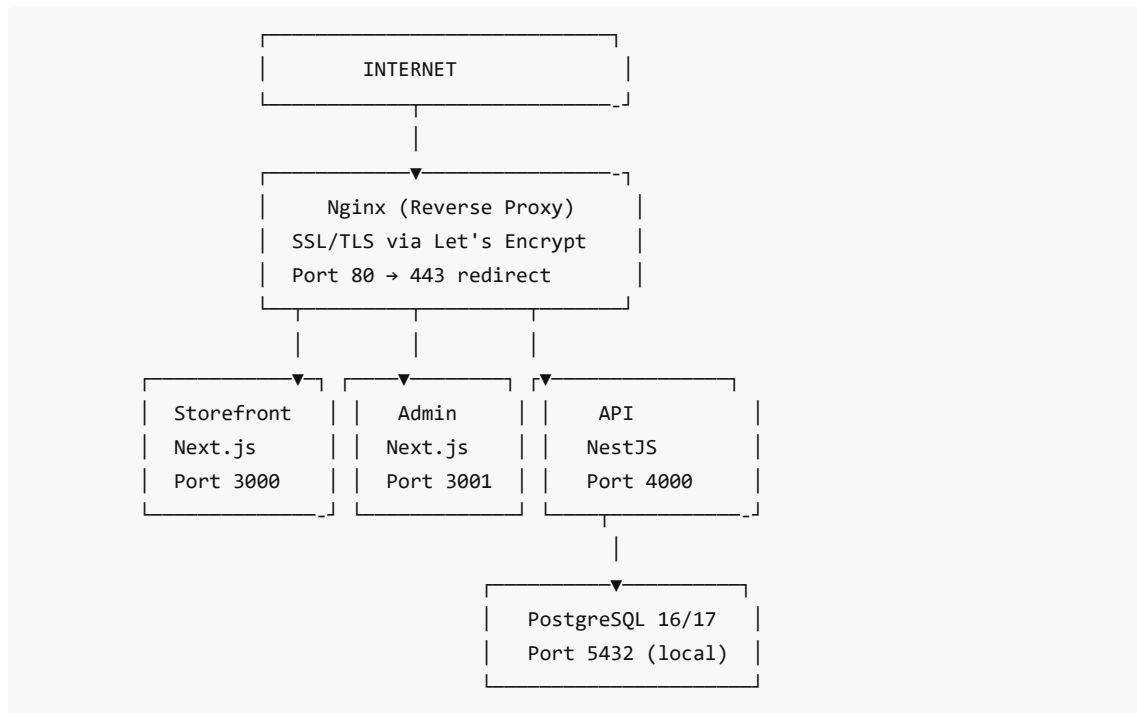


Naro Fashion — Production Deployment Guide

Overview

This guide covers the complete deployment of the Naro Fashion multi-tenant SaaS e-commerce platform from scratch to a production server.

Architecture:



URLs:

- `narofashion.co.tz` → Storefront (customer-facing shop)
- `admin.narofashion.co.tz` → Admin dashboard
- `api.narofashion.co.tz` → REST API

Prerequisites

- A domain name (e.g., `narofashion.co.tz`)
- A VPS with at least 2GB RAM (Vultr, Hetzner, DigitalOcean, Linode)
- A GitHub account with the repo: `github.com/rjuriio/naro-fashion`
- A credit/debit card for VPS billing

Step 1: Provision a VPS

Option A: Vultr (Recommended)

1. Go to <https://www.vultr.com> and create an account
2. Click **Deploy +** → **Cloud Compute** → **Shared CPU**
3. Configure:

Setting	Value
Location	Frankfurt, DE (closest to East Africa)
Image	Ubuntu 24.04 LTS
Plan	vhf-1c-2gb (\$12/mo) or vc2-2c-2gb (\$15/mo)
SSH Key	Add your public key (optional)
Hostname	naro-fashion-prod

4. Click **Deploy**

5. Note the **IP address** and **root password** from the server details page

Option B: Hetzner (\$7.50/mo)

1. Go to <https://console.hetzner.com>
2. Requires ID verification (passport upload)
3. Create project → Add Server → Ubuntu 24.04, CX32, Falkenstein

Option C: DigitalOcean (\$12/mo)

1. Go to <https://www.digitalocean.com>
2. Create Droplet → Ubuntu 24.04, Basic, Regular \$12/mo

Step 2: Connect to the Server

Generate SSH Key (if you don't have one)

```
# On your local machine (Git Bash, Mac Terminal, or Linux)
ssh-keygen -t ed25519 -C "your@email.com"
cat ~/.ssh/id_ed25519.pub
# Copy the output and add to server
```

Connect via SSH

```
ssh root@<YOUR_VPS_IP>
# Enter password if no SSH key was added
```

Add SSH Key to Server (if using password login)

```
# On the server:
echo "ssh-ed25519 AAAA... your@email.com" >> ~/.ssh/authorized_keys
```

Step 3: Install Server Dependencies

Run all commands on the server as root:

```
# Update system
apt update && apt upgrade -y

# Install essentials
apt install -y curl git build-essential nginx certbot python3-certbot-nginx ufw

# Install Node.js 22 LTS
curl -fsSL https://deb.nodesource.com/setup_22.x | bash -
apt install -y nodejs

# Install pnpm (package manager)
npm install -g pnpm@10

# Install PM2 (process manager)
npm install -g pm2

# Install PostgreSQL
apt install -y postgresql postgresql-contrib
systemctl enable postgresql
systemctl start postgresql

# Verify installations
node -v      # Should show v22.x
pnpm -v     # Should show 10.x
pm2 -v      # Should show 6.x
psql --version # Should show 16.x
```

Step 4: Configure PostgreSQL

```
# Generate a strong password
openssl rand -base64 32

# Create database and user
sudo -u postgres psql
```

Inside PostgreSQL:

```
CREATE USER naro_admin WITH PASSWORD '<YOUR_STRONG_PASSWORD>';
CREATE DATABASE naro_fashion OWNER naro_admin;
GRANT ALL PRIVILEGES ON DATABASE naro_fashion TO naro_admin;
\q
```

Save the password — you'll need it for the `.env` file.

Step 5: Clone the Repository

```
mkdir -p /var/www/naro-fashion
cd /var/www/naro-fashion
git clone https://github.com/rjurio/naro-fashion.git .
```

Step 6: Configure Environment Variables

Main `.env` file

```
cat > /var/www/naro-fashion/.env << 'EOF'
NODE_ENV=production

DATABASE_URL="postgresql://naro_admin:<DB_PASSWORD>@localhost:5432/naro_fashion?
schema=public"

JWT_SECRET="<GENERATE: openssl rand -base64 64>"
JWT_REFRESH_SECRET="<GENERATE: openssl rand -base64 64>"
JWT_EXPIRATION="15m"
JWT_REFRESH_EXPIRATION="7d"

STOREFRONT_URL="https://narofashion.co.tz"
ADMIN_URL="https://admin.narofashion.co.tz"
API_URL="https://api.narofashion.co.tz"
NEXT_PUBLIC_API_URL="https://api.narofashion.co.tz/api/v1"

SMTP_HOST=""
SMTP_PORT="587"
SMTP_USER=""
SMTP_PASS=""
SMTP_FROM="noreply@narofashion.co.tz"

INSTAGRAM_BUSINESS_ACCOUNT_ID="17841418108905851"
FACEBOOK_APP_ID="4338851449722487"
EOF
```

Replace `<DB_PASSWORD>` with your PostgreSQL password, and generate JWT secrets:

```
openssl rand -base64 64 | tr -dc 'a-zA-Z0-9' | head -c 64
```

Copy env to sub-packages

```
cp .env packages/database/.env
cp .env apps/api/.env

# Storefront env
echo 'NEXT_PUBLIC_API_URL=https://api.narofashion.co.tz/api/v1' > apps/storefront/.env.local
echo 'NEXT_PUBLIC_TENANT_SLUG=naro-fashion' >> apps/storefront/.env.local
```

```
# Admin env
echo 'NEXT_PUBLIC_API_URL=https://api.narofashion.co.tz/api/v1' > apps/admin/.env.local
```

Step 7: Install Dependencies & Build

```
cd /var/www/naro-fashion

# Install all dependencies
pnpm install

# Install multer (required for file uploads)
pnpm add multer @types/multer --filter api

# Generate Prisma client
cd packages/database && npx prisma generate && cd ../..

# Push schema to database
cd packages/database && npx prisma db push --accept-data-loss && cd ../..

# Run multi-tenant migration (creates first tenant + platform admin)
node packages/database/prisma/migrate-to-multi-tenant.js

# Seed tenant data (settings, CMS pages, payment methods)
node packages/database/prisma/seed-tenant.js

# Build all 3 apps
pnpm build
```

Expected build output: Tasks: 3 successful, 3 total

If build fails with prerender errors, add `export const dynamic = "force-dynamic";` to the root layout of both Next.js apps.

Step 8: Configure PM2 (Process Manager)

The `ecosystem.config.js` file is already in the repo. For standalone builds:

```
cat > /var/www/naro-fashion/ecosystem.config.js << 'EOF'
module.exports = {
  apps: [
    {
      name: 'naro-api',
      cwd: '/var/www/naro-fashion/apps/api',
      script: 'dist/main.js',
      instances: 1,
      env: { NODE_ENV: 'production', PORT: 4000 },
    },
    {
      name: 'naro-storefront',
```

```

    cwd: '/var/www/naro-fashion/apps/storefront/.next/standalone/apps/storefront',
    script: 'server.js',
    instances: 1,
    env: { NODE_ENV: 'production', PORT: 3000, HOSTNAME: '0.0.0.0' },
  },
  {
    name: 'naro-admin',
    cwd: '/var/www/naro-fashion/apps/admin/.next/standalone/apps/admin',
    script: 'server.js',
    instances: 1,
    env: { NODE_ENV: 'production', PORT: 3001, HOSTNAME: '0.0.0.0' },
  },
],
};
EOF

```

Copy static assets for standalone mode

```

# Storefront static files
cp -r apps/storefront/.next/static
apps/storefront/.next/standalone/apps/storefront/.next/static
cp -r apps/storefront/public apps/storefront/.next/standalone/apps/storefront/public

# Admin static files
cp -r apps/admin/.next/static apps/admin/.next/standalone/apps/admin/.next/static
cp -r apps/admin/public apps/admin/.next/standalone/apps/admin/public

```

Start all apps

```

cd /var/www/naro-fashion
pm2 start ecosystem.config.js
pm2 save
pm2 startup    # Follow the printed command to enable auto-start

```

Verify all apps are running

```

pm2 list
# All 3 should show "online" status

```

Step 9: Configure Nginx

```

cat > /etc/nginx/sites-available/narofashion << 'NGINX'
# Storefront
server {
    listen 80;
    server_name narofashion.co.tz www.narofashion.co.tz;

```

```

location / {
    proxy_pass http://127.0.0.1:3000;
    proxy_http_version 1.1;
    proxy_set_header Upgrade $http_upgrade;
    proxy_set_header Connection 'upgrade';
    proxy_set_header Host $host;
    proxy_set_header X-Real-IP $remote_addr;
    proxy_set_header X-Forwarded-For $proxy_add_x_forwarded_for;
    proxy_set_header X-Forwarded-Proto $scheme;
    proxy_cache_bypass $http_upgrade;
}

}

# Admin
server {
    listen 80;
    server_name admin.narofashion.co.tz;
    location / {
        proxy_pass http://127.0.0.1:3001;
        proxy_http_version 1.1;
        proxy_set_header Upgrade $http_upgrade;
        proxy_set_header Connection 'upgrade';
        proxy_set_header Host $host;
        proxy_set_header X-Real-IP $remote_addr;
        proxy_set_header X-Forwarded-For $proxy_add_x_forwarded_for;
        proxy_set_header X-Forwarded-Proto $scheme;
        proxy_cache_bypass $http_upgrade;
    }
}

# API
server {
    listen 80;
    server_name api.narofashion.co.tz;
    client_max_body_size 30M;
    location / {
        proxy_pass http://127.0.0.1:4000;
        proxy_http_version 1.1;
        proxy_set_header Upgrade $http_upgrade;
        proxy_set_header Connection 'upgrade';
        proxy_set_header Host $host;
        proxy_set_header X-Real-IP $remote_addr;
        proxy_set_header X-Forwarded-For $proxy_add_x_forwarded_for;
        proxy_set_header X-Forwarded-Proto $scheme;
        proxy_cache_bypass $http_upgrade;
    }
    location /uploads/ {
        proxy_pass http://127.0.0.1:4000/uploads/;
        add_header Cache-Control "public, max-age=604800";
    }
}

NGINX

```

```
# Enable site and test
ln -sf /etc/nginx/sites-available/narofashion /etc/nginx/sites-enabled/
rm -f /etc/nginx/sites-enabled/default
nginx -t
systemctl reload nginx
```

Step 10: Configure DNS (Habari Node)

The domain `narofashion.co.tz` is registered with **Habari Node** (`habarinode.co.tz` / `habari.co.tz`), a Tanzanian ISP and domain registrar accredited by tzNIC.

10a. Login to Habari Node Client Portal

1. Go to <https://hosting.habari.co.tz/>
2. Login with your Habari Node account credentials
3. Navigate to **Domains** → select `narofashion.co.tz`
4. Click **DNS Management** or **Zone Editor** or **Manage DNS**

*If Habari Node uses cPanel/WHM (common for Tanzanian hosts), look for: **Domains** → **Zone Editor** in the cPanel sidebar*

10b. Add DNS A Records

Add these 4 records (delete any existing A records pointing elsewhere first):

Type	Name/Host	Value/Points to	TTL
A	@	80.240.30.107	3600
A	www	80.240.30.107	3600
A	admin	80.240.30.107	3600
A	api	80.240.30.107	3600

Notes:

- @ means the root domain (`narofashion.co.tz`). Some registrars show this as blank or the full domain name
- For `www` , `admin` , `api` — enter just the subdomain name (not `www.narofashion.co.tz`)
- Type must be **A** (not CNAME, not MX)
- If TTL options are limited, choose the lowest available (300, 600, or 3600)

10c. If Habari Node Uses Nameservers Instead of Direct DNS

Some registrars don't provide a DNS zone editor. Instead, you point nameservers to a DNS provider. In that case:

Option A: Use Vultr's DNS (free)

1. In Vultr dashboard → **DNS** → **Add Domain** → enter `narofashion.co.tz`
2. Add the 4 A records above in Vultr's DNS panel
3. In Habari Node, change nameservers to:
 - `ns1.vultr.com`

- ns2.vultr.com

Option B: Use Cloudflare (free)

1. Go to <https://dash.cloudflare.com> → **Add Site** → enter narofashion.co.tz
2. Select **Free** plan
3. Add the 4 A records above in Cloudflare's DNS panel
4. In Habari Node, change nameservers to the ones Cloudflare provides (e.g., xxx.ns.cloudflare.com)

10d. If You Need Help from Habari Node

Contact Habari Node support:

- **Portal:** <https://hosting.habari.co.tz/> (open a support ticket)
- **Email:** software@habari.co.tz
- **Phone:** +255 659 074 444
- **Office:** Arusha, Tanzania

Tell them: "I need to add A records for narofashion.co.tz pointing to IP 80.240.30.107. I need records for @, www, admin, and api subdomains."

10e. Verify DNS Propagation

Wait 5-30 minutes (up to 24 hours for nameserver changes), then test:

```
# From your terminal
dig narofashion.co.tz +short
# Should return: 80.240.30.107

dig admin.narofashion.co.tz +short
# Should return: 80.240.30.107

dig api.narofashion.co.tz +short
# Should return: 80.240.30.107
```

Or use <https://dnschecker.org> to check propagation globally.

Step 11: Install SSL Certificates

After DNS is pointing correctly:

```
certbot --nginx \
  -d narofashion.co.tz \
  -d www.narofashion.co.tz \
  -d admin.narofashion.co.tz \
  -d api.narofashion.co.tz \
  --email hello@narofashion.co.tz \
  --agree-tos \
  --non-interactive

# Verify auto-renewal
certbot renew --dry-run
```

Step 12: Set Tenant Domain

Connect the production domain to the tenant in the database:

```
# Via psql (most reliable method)
sudo -u postgres psql -d naro_fashion -c "UPDATE \"Tenant\" SET \"domain\" =
'narofashion.co.tz' WHERE \"slug\" = 'naro-fashion';"

# Restart storefront so middleware picks up the domain
pm2 restart naro-storefront
```

Note: The `node -e` approach may fail due to `pnpm` module resolution. Using `psql` directly is more reliable on the server.

Step 13: Configure Firewall

```
ufw allow OpenSSH
ufw allow 'Nginx Full'
ufw enable
ufw status
```

Step 14: Set Up Database Backups

```
# Create backup directory
mkdir -p /var/backups

# Add daily backup cron (runs at 2 AM)
crontab -e
# Add these lines:
0 2 * * * pg_dump -U naro_admin naro_fashion | gzip > /var/backups/naro_fashion_$(date
+\\%F).sql.gz
0 3 * * * find /var/backups -name "naro_fashion_*.sql.gz" -mtime +30 -delete
```

Step 15: Set Up Log Management

```
pm2 install pm2-logrotate
pm2 set pm2-logrotate:max_size 10M
pm2 set pm2-logrotate:retain 7
```

Step 16: Add Swap Space (Recommended for 2GB RAM)

```
fallocate -l 2G /swapfile
chmod 600 /swapfile
mkswap /swapfile
swapon /swapfile
echo '/swapfile swap swap defaults 0 0' >> /etc/fstab
free -h # Verify swap is active
```

Post-Deployment Checklist

#	Task	How to Verify
1	Storefront loads	Visit https://narofashion.co.tz
2	Admin loads	Visit https://admin.narofashion.co.tz
3	API responds	Visit https://api.narofashion.co.tz/api/docs
4	SSL working	Check padlock icon in browser
5	Admin login	<code>admin@narofashion.co.tz / admin123</code>
6	Platform login	<code>platform@naro.co.tz / Admin123</code>
7	CHANGE PASSWORDS	Change both admin + platform admin passwords immediately
8	Products visible	Browse storefront, check images load
9	Mobile works	Open site on phone
10	Backups running	<code>ls /var/backups/</code> after midnight
11	Swap enabled	<code>free -h</code> shows swap
12	PM2 auto-start	<code>sudo reboot</code> then verify apps come back

Updating the Application

Manual Update

```
cd /var/www/naro-fashion
git pull origin master
pnpm install
cd packages/database && npx prisma generate && npx prisma db push --accept-data-loss && cd ../..
pnpm build

# Copy static files for standalone mode
cp -r apps/storefront/.next/static
apps/storefront/.next/standalone/apps/storefront/.next/static
cp -r apps/storefront/public apps/storefront/.next/standalone/apps/storefront/public
cp -r apps/admin/.next/static apps/admin/.next/standalone/apps/admin/.next/static
```

```
cp -r apps/admin/public apps/admin/.next/standalone/apps/admin/public

pm2 restart all
```

Using the Deploy Script

```
chmod +x deploy.sh
./deploy.sh
```

Monitoring

```
pm2 monit          # Live CPU/memory monitor
pm2 logs           # Stream all logs
pm2 logs naro-api  # Stream API logs only
pm2 list           # Check status of all apps
pm2 info naro-api  # Detailed info for one app
```

Troubleshooting

App won't start (PM2 shows "errored")

```
pm2 logs <app-name> --lines 50  # Check error logs
```

502 Bad Gateway

- App hasn't started yet — wait 10-15 seconds after restart and retry
- Check `pm2 list` to see if apps are online (status = "online")
- If API keeps restarting (high restart count), check logs: `pm2 logs naro-api --lines 30`

"Cannot find module" errors on API

```
# Common: multer not installed
cd /var/www/naro-fashion && pnpm add multer @types/multer --filter api
pm2 restart naro-api
```

Database connection failed

```
sudo -u postgres psql -c "SELECT 1"  # Test PostgreSQL is running
systemctl status postgresql          # Check service status
systemctl restart postgresql         # Restart if needed
```

Run raw SQL on production database

```
sudo -u postgres psql -d naro_fashion # Opens psql shell
# Then run SQL queries, e.g.:
# SELECT * FROM "Tenant";
# \q to exit
```

SSL certificate issues

```
certbot renew --force-renewal # Force renew certificates
nginx -t && systemctl reload nginx # Reload Nginx
certbot certificates # List all certificates + expiry dates
```

Next.js build fails with prerender errors

Both Next.js apps need these settings in `next.config.js` :

```
typescript: { ignoreBuildErrors: true },
eslint: { ignoreDuringBuilds: true },
output: 'standalone',
```

And root `app/layout.tsx` needs at top:

```
export const dynamic = "force-dynamic";
```

Client pages (`"use client"`) also need `export const dynamic = "force-dynamic";` AFTER the `"use client"` directive.

Standalone build — static files missing (blank pages)

After every `pnpm build` , you must copy static files:

```
cp -r apps/storefront/.next/static
apps/storefront/.next/standalone/apps/storefront/.next/static
cp -r apps/storefront/public apps/storefront/.next/standalone/apps/storefront/public
cp -r apps/admin/.next/static apps/admin/.next/standalone/apps/admin/.next/static
cp -r apps/admin/public apps/admin/.next/standalone/apps/admin/public
```

Out of memory during build

```
# Check memory
free -h

# If no swap, add it:
fallocate -l 2G /swapfile && chmod 600 /swapfile && mkswap /swapfile && swapon /swapfile
echo '/swapfile swap swap defaults 0 0' >> /etc/fstab

# Or build on your local machine and deploy the built files
```

Nginx returns wrong site for a domain

```
nginx -t                                # Test config syntax
cat /etc/nginx/sites-enabled/narofashion # Verify server_name entries
systemctl reload nginx                  # Apply changes
```

Known Build & Deployment Issues

These are issues discovered during the actual deployment:

Issue	Root Cause	Fix
Cannot find module 'multer'	pnpm strict hoisting doesn't resolve multer for API	pnpm add multer @types/multer - -filter api
Module not found: '@/types/model-viewer'	.d.ts files can't be imported as modules	Change import '@/types/...' to /// <reference path="..." />
The "use client" directive must be placed before other expressions	export const dynamic added before "use client"	dynamic export must come AFTER "use client"
Error occurred prerendering page "/auth/login"	Pages use localStorage/browser APIs during SSR prerender	Add export const dynamic = "force-dynamic" to root layout
Property 'imageUrl' does not exist on type 'Category'	TypeScript strict mode catches missing interface fields	Add missing field to interface or disable TS checks in build
Cannot find module '@prisma/client' when running scripts from /tmp	pnpm node_modules not in /tmp resolve path	Run scripts from project directory or use psql for DB queries
Standalone Next.js shows blank page	Static assets not copied to standalone directory	Copy .next/static and public after each build

Infrastructure Reference

Current Production Setup

Component	Details
VPS Provider	Vultr (vultr.com)
Server Plan	vhf-1c-2gb (1 vCPU, 2GB RAM, 64GB NVMe SSD)
Server Location	Frankfurt, Germany
Server IP	80.240.30.107
OS	Ubuntu 24.04 LTS
Domain Registrar	Habari Node (habarinode.co.tz / hosting.habari.co.tz)

DNS Provider	Vultr DNS (ns1.vultr.com, ns2.vultr.com)
SSL	Let's Encrypt (auto-renews, expires 2026-06-18)
Node.js	v22.x LTS
Package Manager	pnpm v10.x
Process Manager	PM2
Web Server	Nginx
Database	PostgreSQL 16
App Directory	/var/www/naro-fashion

Port Mapping

Domain	Nginx →	App
narofashion.co.tz	127.0.0.1:3000	Storefront (Next.js standalone)
admin.narofashion.co.tz	127.0.0.1:3001	Admin (Next.js standalone)
api.narofashion.co.tz	127.0.0.1:4000	API (NestJS dist)

DNS Records (Vultr DNS)

Type	Name	Value	TTL
A	@	80.240.30.107	300
A	www	80.240.30.107	3600
A	admin	80.240.30.107	3600
A	api	80.240.30.107	3600
CNAME	*	narofashion.co.tz	300
NS		ns1.vultr.com	300
NS		ns2.vultr.com	300

Default Credentials (CHANGE IMMEDIATELY)

Login	Email	Password	URL
Tenant Admin	admin@narofashion.co.tz	admin123	https://admin.narofashion.co.tz
Platform Admin	platform@naro.co.tz	Admin123	https://admin.narofashion.co.tz/platform-login

Cost Summary

Service	Monthly Cost	Annual Cost
---------	--------------	-------------

VPS (Vultr vhf-1c-2gb)	\$12.00	\$144.00
Domain (.co.tz via Habari Node)	~\$1.76	TZS 21,186/yr (~\$8.25)
SSL (Let's Encrypt)	Free	Free
Email (Brevo free tier)	Free	Free
DNS (Vultr DNS)	Free	Free
Total	~\$13.76/mo	~\$152.25/yr

Security Hardening

Immediate Actions (after deployment)

1. **Change default passwords** for admin and platform admin
2. **Enable SSH key-only login** — disable password authentication:

```
# Edit SSH config
nano /etc/ssh/sshd_config
# Set: PasswordAuthentication no
# Set: PermitRootLogin prohibit-password
systemctl restart sshd
```

Already Configured

- **Firewall (UFW)**: Only ports 22 (SSH), 80 (HTTP), 443 (HTTPS) open
- **PostgreSQL**: Listens only on localhost (not exposed to internet)
- **Helmet**: CSP, XSS protection, clickjacking prevention enabled on API
- **Rate Limiting**: 100 requests per 60 seconds per IP (NestJS Throttler)
- **Account Lockout**: 5 failed login attempts → 30-minute lock
- **Password Hashing**: bcryptjs with salt round 12
- **JWT Secrets**: 64-byte random strings, unique to this deployment
- **CORS**: Only storefront and admin URLs whitelisted
- **File Upload Limits**: 5MB images, 25MB 3D models

Recommended Future Improvements

- Set up **fail2ban** for SSH brute-force protection
- Add **Cloudflare** as CDN/DDoS protection layer
- Enable **PostgreSQL SSL** connections
- Set up **offsite backup** (e.g., S3-compatible storage)
- Configure **monitoring alerts** (PM2 + uptime monitoring service)

Adding a New Tenant Client

When selling the platform to a new client:

1. **Login** as platform admin at <https://admin.narofashion.co.tz/platform-login>
2. Go to **Tenants** → **New Tenant**
3. Fill in: company name, slug, admin email/password, subscription plan

4. The new tenant gets their own isolated storefront, admin, and data

To connect a **custom domain** for the new tenant:

1. Add their domain as an A record pointing to `80.240.30.107` in Vultr DNS
2. Add SSL: `certbot --nginx -d newclient.co.tz`
3. Update tenant domain: `sudo -u postgres psql -d naro_fashion -c "UPDATE \"Tenant\" SET \"domain\" = 'newclient.co.tz' WHERE \"slug\" = 'new-client-slug';"`
4. Restart storefront: `pm2 restart naro-storefront`

Naro Fashion — Multi-Tenant SaaS E-Commerce Platform Deployed on Vultr VPS • Frankfurt, Germany Server IP: 80.240.30.107 Domain: narofashion.co.tz (registered via Habari Node, DNS via Vultr) Last updated: March 2026